

QUIC-based options for moving real-time encoded media between peers

Presenters:
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15:15–16:15 GMT
TPAC 2026
Second Floor - Room 06
Dublin Ireland

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Breakout session: **Oct 28,** 2026, **15:15-16:15 GMT**
Session info: W3C Calendar
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Joining Link:
https://w3c.zoom.us/j/6920008584?pwd=qLb3hRiudbJz3E2L6FiDeg0Sb1zYUN.1

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Breakout Agenda

The W3C WebTransport working group recently extended its charter to consider ***"...incubating mechanisms for peer-to-peer capability"***.
To this end, we want to gague interest in QUIC-based options for moving real-time data between peers.

We'll present two ideas. The first is leveraging the WebRTC handshake to obtain a p2p connection, prior to establishing a WebTransport session over it.

The second is a presentation on ***"A p2p Vision for QUIC"***, which instead offers a QUIC-native approach of migrating a client-server connection to a peer.

Q&A at the end

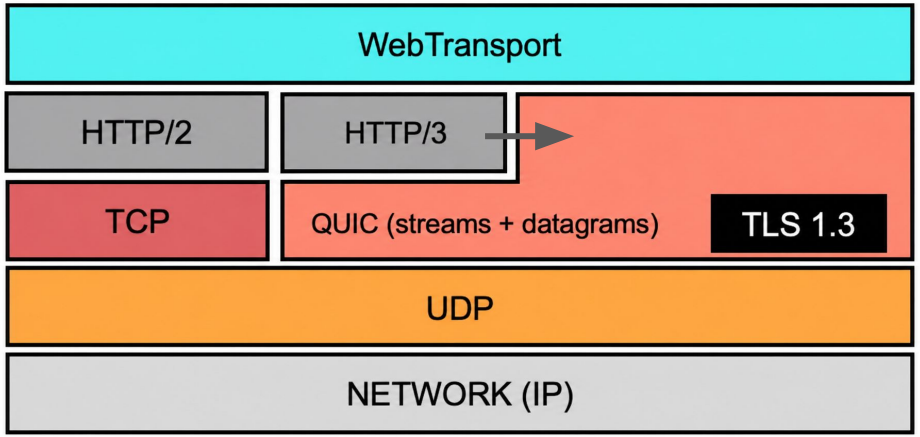
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What is WebTransport?

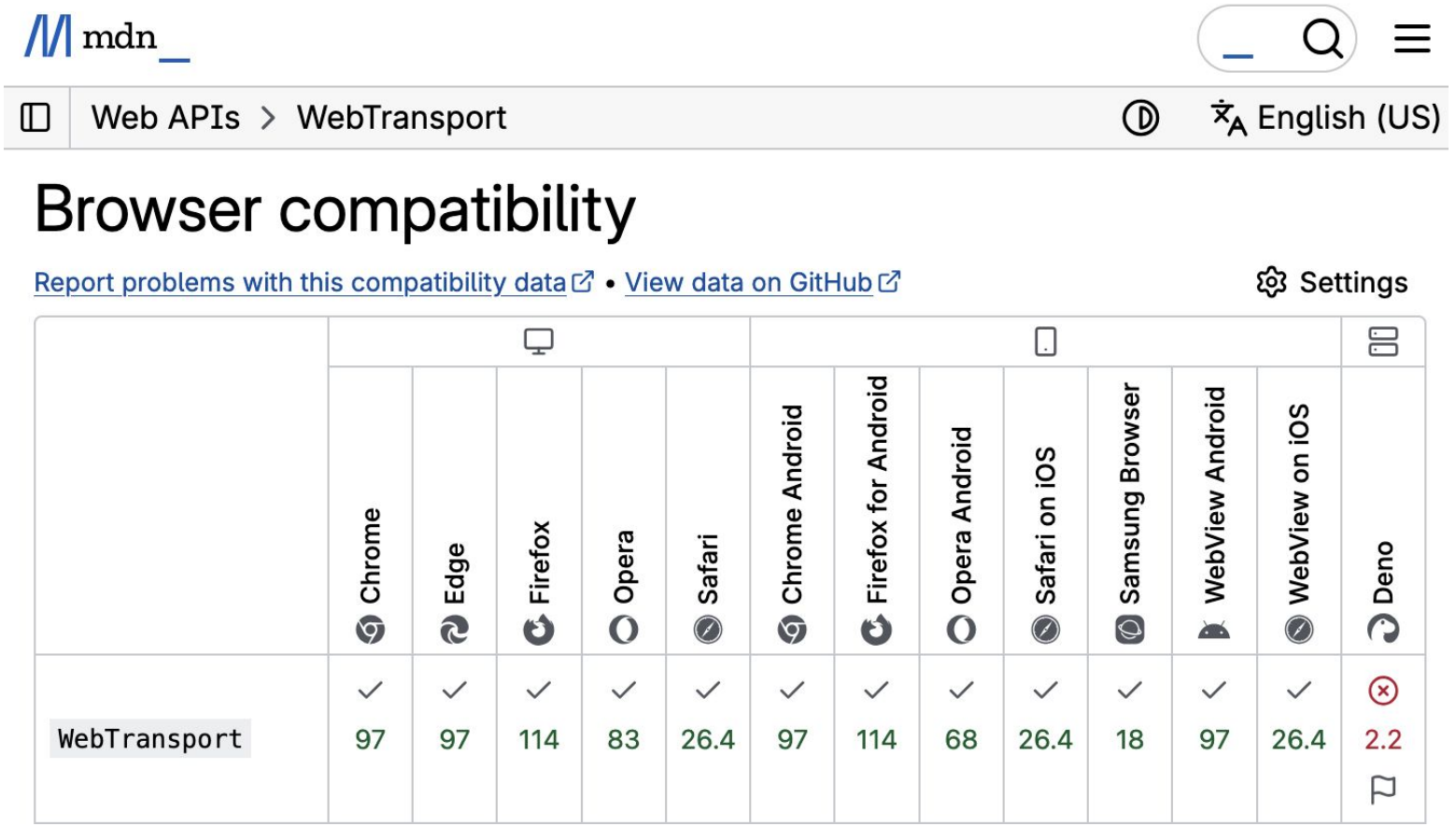
It's a new web platform networking API

When established over HTTP/3, it runs directly over QUIC after the handshake

Client-server, though QUIC streams and datagrams are surprisingly symmetrical



WebTransport API is available in all browsers



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Why consider WebTransport support for peer-to-peer?

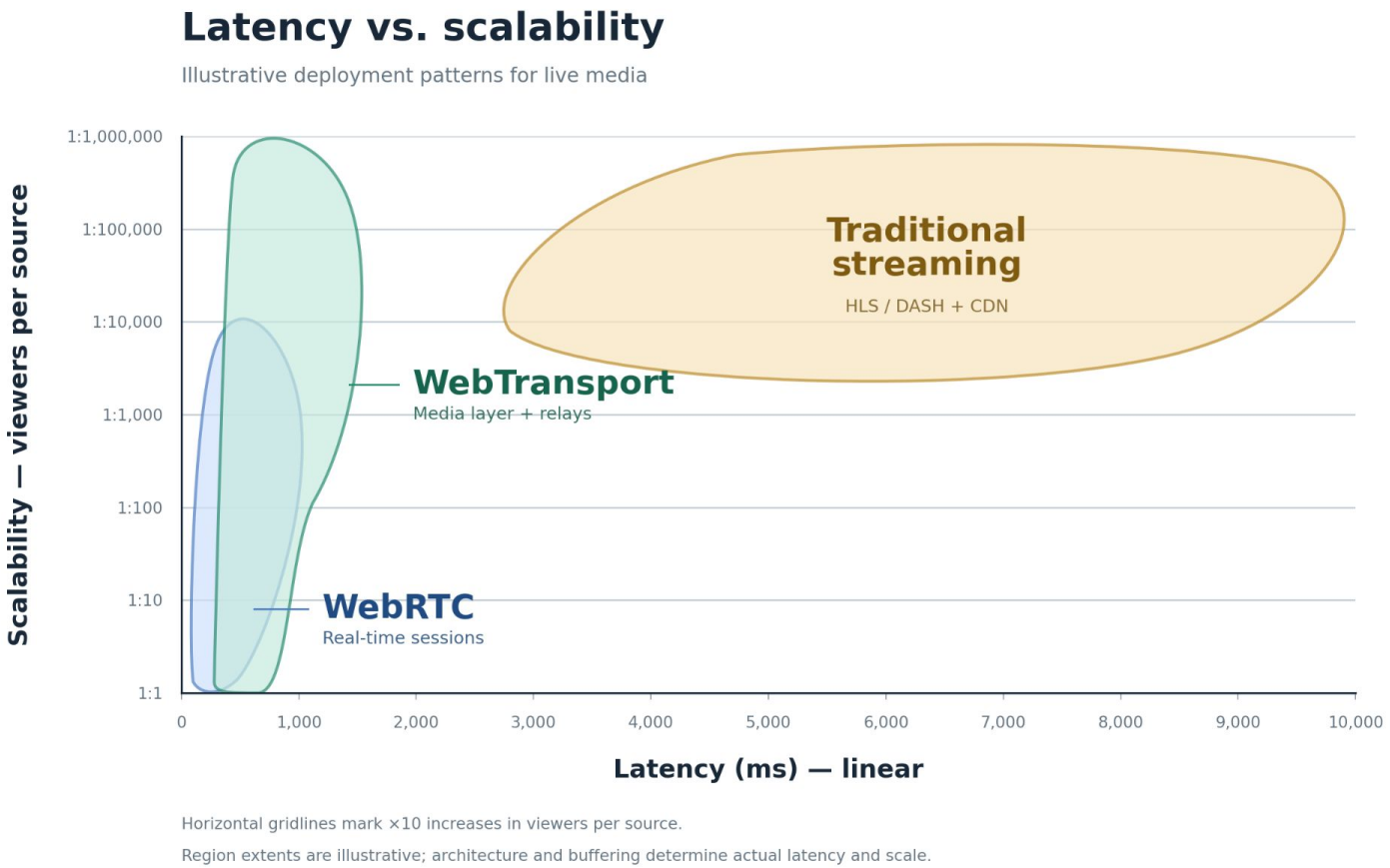
Why?

- QUIC over LAN use cases (MOQ)
- **Long-term:** Potentially simpler alternative to WebRTC
  - Ironically, most (but not all) WebRTC calls today are to a server
  - Close a request gap

Why now?

- Long requested in https://github.com/w3c/webtransport/issues/590
- WG decided to defer it to focus on client-server for 1.0.
- 1.0. is now a Candidate Recommendation.

Overlap with WebRTC (illustration only)



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How to establish a connection with WebTransport (2 RTT)

```
const wt = new WebTransport("https://example.com/wt");
```

How to establish a connection with WebRTC (6 RTT)

```
const signaling = new WebSocket("wss://signaling.example.org");
const iceServers = [{urls: "stun:stun.example.org"}];
const pc = new RTCPeerConnection({iceServers});
await new Promise(r => signaling.onopen = r);
pc.createDataChannel("data");
pc.onicecandidate = e => signaling.send(JSON.stringify(e));
pc.onnegotiationneeded = async () => {
  await pc.setLocalDescription();
  signaling.send(JSON.stringify({description: pc.localDescription}));
};
signaling.onmessage = async ({data}) => {
  const {description, candidate} = JSON.parse(data);
  if (description) {
    await pc.setRemoteDescription(description);
    if (description.type == 'offer') {
      await pc.setLocalDescription();
      signaling.send(JSON.stringify({description: pc.localDescription}));
    }
  } else if (candidate) {
    await pc.addIceCandidate(candidate);
  }
};
```

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Direction of interest? Embrace or replace WebRTC?

Embrace WebRTC

- *"WebRTC is a good/working standard solution for P2P"*
- Piggyback on WebRTC's connection establishment
  - WebRTC tries for p2p first, with fallback to relay
- Developer group: love/already familiar with WebRTC
- Quick and dirty path. s/createDataChannel/createWebTransport/

Replace WebRTC

- *"I want WebTransport to solve P2P establishment, without WebRTC"*
- Solve it in QUIC with a relay server and path migration
  - Relay by default, promotes to p2p
- Developer group: No fan/unfamiliar with WebRTC
- Potential for cleaner solution independent of WebRTC

QUIC (and dirty) over WebRTC ICE

From napkin [https://github.com/w3c/webtransport/issues/590 (comment)](https://github.com/w3c/webtransport/issues/590):

```
const pc = new RTCPeerConnection();
// ...webrtc signaling from earlier slide collapsed here
const wt = pc.createWebTransport(options);
await wt.ready;
```

Developer gets familiar WebTransport surface, just not from constructor (or fetch)

- JSEP would need to describe how to negotiate QUIC over ICE in SDP
- We'd need to define WebTransport handshake happening over ICE transport
- If WebRTC falls back to relay we just wasted 6 RTT
- Unanswered questions: Will it work? Can it co-exist with RTCRtpTransceivers?
- How are connection drops handled? Awkward API

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Presentation #2 (Marten)

https://marten-seemann.github.io/tpac-2026-quic-nat-traversal

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Questions and feedback

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End of Slides